# Orbit-Consistent Tesseract Trees on Frozen Foundation Features for Spacecraft Pose Estimation across the Domain Gap

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## Abstract

*Monocular spacecraft pose estimators trained on synthetic renders lose much of their accuracy on real hardware-in-the-loop images. We study this gap on SPEED+ with a deliberately lightweight design: two frozen foundation backbones (DINOv3 for appearance, MoGe-2 for surface geometry) feed small autoregressive Transformers that predict a rotation as a root-to-leaf path in a hierarchical tesseract partition of $SO(3)$, and the two branches are fused per tree level by a training-free product of experts. An audit of the remaining real-domain errors shows that they are dominated by $180°$ flips about body axes of a nearly symmetric spacecraft, and that the correct mode is usually already among the predictor's hypotheses: the problem is* selection*, not representation. We resolve most of it with* orbit voting*: the crop is rotated in-plane four times, every view's hypotheses are mapped back through an exactly derived camera-roll transform, and candidates are scored by their tree likelihood in all views. Combined with photometric augmentation, this reduces the mean rotation error from $26.9°/36.7°$ to $12.3°/12.3°$ on the lightbox/sunlamp domains while training only 2.4 M parameters per branch. Distilling the orbit system into single-view students further lowers it to $11.2°/11.1°$, and a translation block on the same tokens extends the model to 6-DoF without hurting rotation. In pre-registered ablations, the tesseract tree is far more accurate than a count-matched flat classifier, and photometric augmentation, not rotation augmentation, carries the real-domain gain. Orbit voting also transfers to T-LESS and SYMSOL-T, although absolute accuracy there remains below fine-tuned state-of-the-art methods. We report negative results throughout.*

## 1. Introduction

Estimating the pose of an uncooperative spacecraft from a single image is a prerequisite for autonomous rendezvous, servicing and debris removal. Because real orbital imagery with ground truth is scarce, models are trained on synthetic renders and must transfer to real images with harsh, directional illumination. SPEED+ [18] makes this gap measurable: models train on synthetic images and are tested on hardware-in-the-loop *lightbox* and *sunlamp* captures of a physical mock-up. Even strong pipelines lose much of their accuracy there [4, 16].

We approach the problem from the side of *frozen* foundation models. DINOv3 [22] gives robust appearance features, and MoGe-2 [23] gives per-pixel surface normals. If such features already carry the pose information, a small head trained on synthetic data could transfer better than a fine-tuned network that can over-fit to render statistics. Our head predicts a rotation as a path in a hierarchical partition of $SO(3)$ derived from the recursive tesseract subdivision [14]. It is an autoregressive Transformer that first chooses one of 4 charts, then one of 8 children at each of 5 levels (131,072 cells), attending to the frozen tokens. The DINOv3 and MoGe-2 trees are fused per level by a training-free product of experts (PoE) [8].

This baseline is accurate on synthetic data ($5.5°$) but not on real images ($26.9°$ lightbox, $36.7°$ sunlamp). A careful audit (Sec. 4.2) shows that the real-domain error is not spread over all rotations. It is concentrated in $180°$ flips about a few body axes of the nearly symmetric spacecraft, at about twice the rate expected for random errors. Crucially, the correct mode is usually present among the predictor's own hypotheses: an oracle that picks the best of the tree's candidates and their flips reaches $9.8°$ on lightbox. The failure is one of *selection*. Continuous $SO(3)$ flow posteriors, flow-based re-scoring, template matching and discrete path flows did not fix it (Appendix C).

What does fix most of it is a simple, geometry-exact form of test-time consistency that we call *orbit voting* (Sec. 3.4). Rotating the crop in the image plane is equivalent to rolling the camera about the ray through the crop centre, so the true rotation transforms by a known left multiplication. A wrong look-alike pose tends to win only in some of the rotated views, whereas the true pose, mapped back, is consistent across all of them. We therefore collect the tree's beam

candidates from four rotated views, map them to a common frame, and score each candidate by its tree likelihood as seen from every view. Training with photometric augmentation and in-plane rotations, labelled through the same transform, and then applying orbit voting reduces the error to 12.3° on both real domains.

Our contributions are:

- A lightweight hierarchical rotation predictor on frozen DINOv3 and MoGe-2 features, with training-free per-level PoE fusion. In a matched comparison it is far more accurate than a flat 131k-anchor Hopf classifier of the kind used in recent spacecraft work [4], with 15× fewer parameters (Sec. 5).
- A diagnosis that real-domain error on SPEED+ is dominated by flip *mode selection*, with the correct mode usually among the hypotheses (Sec. 4.2).
- Orbit voting with an exactly derived roll transform for hierarchical candidates. With photometric augmentation it repairs 62–80% of severe flips while breaking 0.4% of good predictions. Orbit distillation into single-view students further reduces the error to 11.2°/11.1° (Secs. 3.4 and 3.5).
- A pre-registered evaluation protocol with three seeds, selection on synthetic validation only, and full reporting of negative results. It includes the evidence that photometric augmentation, not rotation augmentation, carries the real-domain gain (Sec. 5). It also covers a 6-DoF extension (Sec. 4.4) and transfer to T-LESS, SYMSOL-T and YCB-V (Sec. 4.5).

Augmentation, test-time augmentation and self-distillation are established techniques. Our claim is about their exact geometric form for hierarchical rotation hypotheses and their measured effect on a diagnosed failure mode, not about the techniques themselves.

## 2. Related Work

**Spacecraft pose estimation.** Early learning-based methods regress or classify pose directly [21] or detect keypoints and solve PnP [17]. SPEED+ [18] shifted the focus to the synthetic-to-real gap. SPNv2 [16] combines multi-task learning, heavy style augmentation and online refinement, and adversarial or self-training adaptation has also been used [19]. Direct regression with Vision Transformers is fast [1]. SPACE-HOP [4] pre-trains a ViT with a joint-embedding predictive objective and classifies rotation over a Hopf-fibration grid with tangent-space offsets. It reports a characteristic W-shaped error histogram with a secondary peak at 180°, which we analyse as a mode-selection problem. Unlike these works, we keep the backbones frozen and train only small heads.

**Discrete and distributional rotation prediction.** Uniform $SO(3)$ grids can be built from the Hopf fibration [25] or by recursive tesseract subdivision [14]; we adapt the latter to cell-centre representatives with unique paths. Grid- and implicit-density methods model multi-modal rotation posteriors, either through MLP scores over grids [11, 15] or through equivariant spherical features [13]; parametric alternatives include Bingham mixtures [3]. SpyroPose [6] evaluates a hierarchical $SE(3)$ pyramid sparsely, coarse to fine. LiePose [12] applies score-based diffusion on $SE(3)$ and handles symmetric objects well. Continuous rotation regression has known representation issues [26]. Riemannian [2] and discrete [5] flow matching provide generative alternatives; we test both as refiners and report them as negative results.

**Consistency, test-time augmentation and distillation.** Averaging predictions over transformed inputs is a standard way to obtain test-time robustness. Data distillation [20] turns such multi-transform ensembles into training targets, a form of knowledge distillation [7]. Equivariant architectures build rotation consistency into the network [13]. Our orbit voting is a test-time augmentation over in-plane rotations, and our orbit distillation is a data-distillation variant. What is specific here is the exact camera-roll transform for crops, scoring with hierarchical tree likelihoods rather than averaging point estimates, and the target: a diagnosed flip-selection failure under domain shift.

## 3. Method

### 3.1. Problem setup

Given a crop of the target around its (ground-truth) bounding box, we predict the rotation $q \in SO(3)$ (unit quaternion, $q \equiv -q$) of the body frame relative to the camera and, in Sec. 4.4, the translation $t \in \mathbb{R}^3$. Rotation error is $E_R = 2\arccos|\langle \hat{q}, q\rangle|$ in degrees, and translation error is $E_T = \|\hat{t} - t\|_2$ in metres. These are the ESA challenge and SPACE-HOP definitions [4]. Crops are square boxes around the projected keypoints with a 15% margin per side.

### 3.2. Tesseract tree on $SO(3)$

Following the recursive tesseract subdivision [14], a quaternion is assigned to a *chart* $c = \arg\max_i |q_i|$, sign-normalised so that $q_c > 0$ (this identifies $q$ and $-q$), and projected to the cube $x = q/q_c$. Its three free coordinates in $[-1, 1]$ are halved recursively, so each level selects one of 8 children. We use *cell centres* (lifted back to the unit sphere) as representatives. Unlike corner representatives, every rotation then has a unique root-to-leaf path. With 4 charts and depth 5 there are $4 \cdot 8^5 = 131{,}072$ cells. The quantisation error of the leaf centre is 2.51° on average (3.97° p95, 6.20° worst case), and it halves per level. This floor bounds the fine accuracy of all our rotation results.

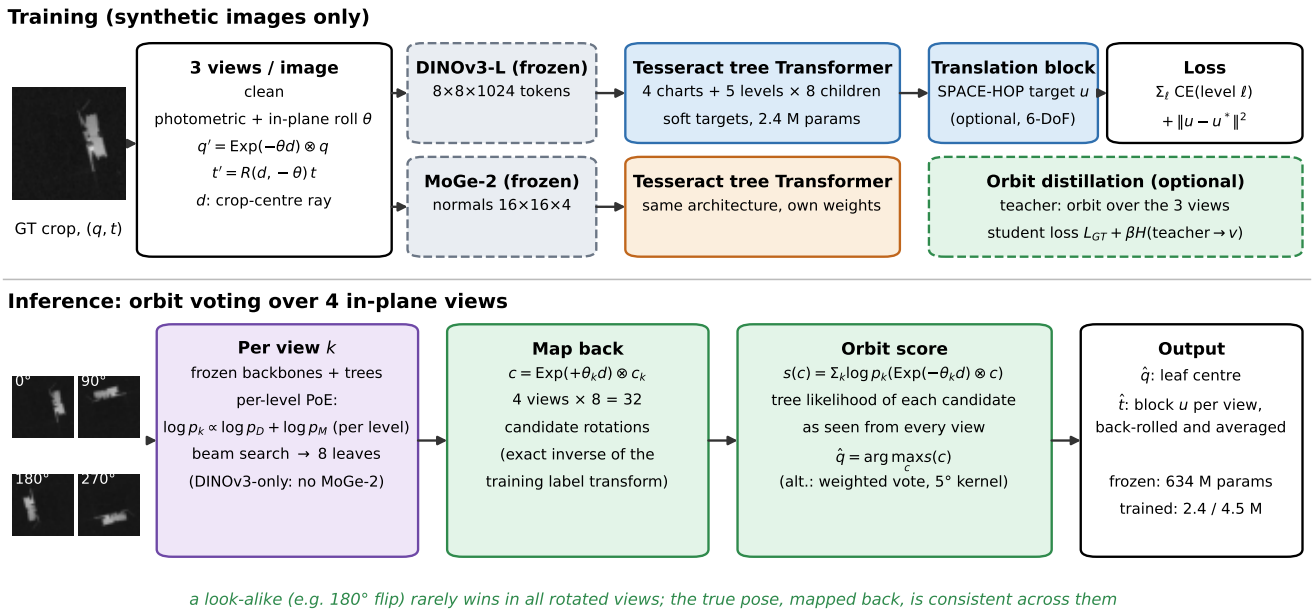



Figure 1. **Overview.** *Top:* training. Each synthetic crop is used as a clean view and as two augmented copies (photometric augmentation plus an in-plane roll $\theta$ with exactly transformed labels). Frozen DINOv3 and MoGe-2 features feed one tesseract tree Transformer each; an optional translation block reads the same tokens, and optional orbit distillation replaces the ground-truth target by an orbit teacher. *Bottom:* orbit voting at test time. Candidates from four rotated views are mapped back to a common frame and scored by their per-level PoE tree likelihood in every view.

**Tree Transformer.** For each backbone, a small decoder predicts the path autoregressively. The frozen tokens are layer-normalised, linearly projected to $d = 256$, given 2-D positional encodings and passed through one encoder layer. A causal decoder with 2 layers (4 heads, FFN 512) then reads a sequence that starts with a root token followed by one token per decided level. Each such token embeds the level and a 6-frequency Fourier encoding of the current cell centre, and cross-attends to the encoded image tokens. The root position predicts the chart (4 logits) and position $\ell$ predicts the child (8 logits). The model has 2.4 M parameters. We train with geodesic soft targets: at each level, the target over the candidate cells is $\propto \exp(-(\delta/\tau_\ell)^2)$, where $\delta$ is the angle between the ground truth and each cell centre. The widths $\tau_\ell$ (39.3° for the root and level 1, 20.0°, 10.0°, 5.0°, 2.5°) halve with the cell size. Training uses teacher forcing on the ground-truth path and a cross-entropy at every level; inference is greedy or beam search, and the prediction is the leaf centre.

**Frozen features.** DINOv3 ViT-L/16 [22] last-layer patch tokens are average-pooled to an $8\times8$ grid ($64 \times 1024$). For geometry, MoGe-2 [23] surface normals are masked by a foreground mask on which both backbones agree (the intersection of a DINOv3 foreground estimate and the MoGe-2 mask). They are then pooled to $16\times16$ cells, each holding three masked-normal channels and the mask fraction ($256 \times 4$). This consensus mask removes most of MoGe-2's background leakage on real images (Appendix B). Features are extracted once; only the trees are trained.

**Per-level product of experts.** The two trees are fused without training. At every decoding step, the log-softmax distributions of both branches are added and renormalised: $\log p^{(\ell)} \propto \log p^{(\ell)}_{\text{DINO}} + \log p^{(\ell)}_{\text{MoGe}}$. Beam search then runs on the fused tree. We report this *PoE* variant and a faster *DINOv3-only* variant.

### 3.3. Augmentation with exact in-plane labels

Each synthetic training image is used as a clean crop and as two pre-extracted augmented copies; one of the three is drawn at every step. The augmentations are photometric (brightness, contrast, gamma, blur, glare blobs, noise, random erasing, JPEG) plus an in-plane rotation of the crop by $\theta \sim \mathcal{U}(-180°, 180°)$ about its centre. Rotating the crop about its centre is equivalent to rotating the camera about the unit ray $d$ through the crop centre, computed with the lens distortion removed. Points therefore transform as $p' = R(d, -\theta)\, p$, and the labels as

$$q' = \text{Exp}(-\theta\, d) \otimes q, \qquad t' = R(d, -\theta)\, t. \tag{1}$$

We derived this convention from the camera geometry rather than fitting it to a predictor. Projecting model keypoints with $q'$ reproduces the warped image to $0.2°$ (p95 label error measured by PnP). For datasets whose objects lie far from the optical axis (T-LESS, YCB-V), an affine crop rotation is not exactly a camera roll (label error up to $6.7°$). There we warp with the roll homography $KR(d,-\theta)K^{-1}$ instead, which makes Eq. (1) exact (verified to $0.000°$).

### 3.4. Orbit voting

At test time we form $K{=}4$ views of the crop rotated by $\theta_k \in \{0°, 90°, 180°, 270°\}$. In view $k$ the fused tree's beam search returns 8 leaves $c_{k,1..8}$, which are mapped back to the original frame by the inverse of Eq. (1), $c = \mathrm{Exp}(+\theta_k d) \otimes c_k$, giving 32 candidates. Each candidate $c$ is then scored by the tree likelihood of its own path when mapped into every view:

$$s(c) = \sum_{k=1}^{K} \log p_k\big(\mathrm{path}(\mathrm{Exp}(-\theta_k d) \otimes c)\big), \qquad (2)$$

and predict $\hat{q} = \arg\max_c s(c)$. Here $\log p_k$ sums the fused root and child log-probabilities along the path, evaluated with teacher forcing. A $180°$ look-alike that wins in one view must also be likely in the other three to be selected, whereas the true pose is consistent by construction. We also evaluate a *weighted vote*, in which each candidate collects Gaussian-kernel support ($5°$) from the other candidates, weighted by their normalised beam probabilities within their view. Orbit voting has no parameters and multiplies inference cost by about $4\times$ (Tab. 1).

### 3.5. Orbit distillation

Orbit voting is more accurate than a single view, so we use it as a teacher. For every training image we run the frozen orbit system over its three available views (clean and two augmented, with known rolls). This gives 24 candidates in the clean frame with weights $w_j = \mathrm{softmax}_j\, s(c_j)$. A new tree per branch is then trained from scratch with the same recipe and the loss

$$\mathcal{L} = \mathcal{L}_{\mathrm{GT}}(v) - \frac{\beta}{L{+}1} \sum_j w_j \sum_{\ell=0}^{L} \log p^{(\ell)}\big(\mathrm{pre}_\ell(\mathrm{Exp}(-\theta_v d) \otimes c_j)\big), \qquad (3)$$

with $\beta = 1$ and $\mathrm{pre}_\ell$ the length-$\ell$ path prefix. Here $v$ is the view drawn for the image, and the teacher candidates are transported into its frame. The student is a single-view model with unchanged inference cost; at test time it can be used alone or with orbit voting. This is a data-distillation scheme [20] applied to hierarchical rotation targets.

### 3.6. Translation block (6-DoF)

For translation we keep the SPACE-HOP parameterisation [4] unchanged. With box centre $(b_x, b_y)$ and size $(b_w, b_h)$, image size $(W, H)$, intrinsics $(f_x, f_y, c_x, c_y)$ and $\alpha = 1.6$, the target is $u = (U_x, U_y, U_z)$ with

$$U_x = \tfrac{1}{b_w}\big(f_x \tfrac{X}{Z} + c_x - b_x\big), \quad U_y = \tfrac{1}{b_h}\big(f_y \tfrac{Y}{Z} + c_y - b_y\big), \qquad (4)$$

$$U_z = \tfrac{1}{2}\big(\tfrac{\alpha b_w}{W} + \tfrac{b_h}{H}\big) Z, \qquad (5)$$

trained with $\|u - u^*\|^2$ and inverted in closed form. A translation block (one learned query cross-attending to the tree's encoded tokens, then an MLP) predicts $u$. It is trained either jointly with the tree ($\mathcal{L}_{\text{tree}} + \mathcal{L}_{\text{trans}}$) or on a frozen tree. Rolled views use $t'$ from Eq. (1) with the unchanged box. With orbit voting, each view's $\hat{t}$ is rolled back and the four are averaged.

## 4. Experiments

### 4.1. Setup and protocol

**Data.** SPEED+ [18] provides 47,966 synthetic training images, 11,994 synthetic validation images, and two real test domains: *lightbox* (6,740) and *sunlamp* (2,791). We train on synthetic training images only.

**Protocol.** Every experiment was pre-registered before it was run: its question, arms, controls and pass criterion were fixed in advance, and protocol changes were recorded before the affected runs. We train 3 seeds and report mean $\pm$ seed standard deviation. A criterion "passes" if the difference exceeds the larger seed standard deviation. Checkpoints, hyper-parameters and variants are selected on synthetic validation only; lightbox and sunlamp are used only for testing. Some ablations also report a *rotated validation* proxy, the synthetic validation set with random in-plane rolls. Failed criteria and negative results are reported (Appendix C).

**Implementation.** Trees are trained for 100 epochs with batch 256, AdamW (learning rate $10^{-3}$, weight decay 0.05, one-cycle schedule) and bf16, which takes about 13 minutes per DINOv3 tree on one GPU. Timings are measured end-to-end at batch 1 on a shared NVIDIA RTX 6000 Ada; other jobs were running on it, so they are upper bounds. All evaluations use ground-truth crops (Sec. 6).

### 4.2. Diagnosis: real-domain errors are flips

Before changing the model, we audited the large real-domain errors of the clean (non-augmented) trees. Encoder and decoder round trips are exact, and the metric, rotation convention and data pipeline were ruled out. Chart seams

were ruled out as well: flat Hopf-grid heads on the same features flip as often or more. Of the errors above 90°, 36–41% lie within 15° of 180°, against 20% for uniformly random wrong rotations. Their body-frame rotation axes cluster around a few axes of the near-symmetric spacecraft body. Allowing a 180° rotation about one body axis lowers the lightbox mean of the DINOv3 tree from 38.9° to 23.5°. The correct mode is usually available: among the tree's beam candidates and their 180° flips, an oracle achieves 9.8° on lightbox, against 26.9° for the fused greedy prediction. Neither a continuous flow posterior, flow-based re-noise scoring, synthetic templates nor a discrete path flow selects it on real images (Appendix C). The problem is *mode selection* under domain shift.

### 4.3. Main results on SPEED+

Table 1 summarises the progression. Augmentation alone (photometric plus roll, single view) cuts the real-domain error of the PoE system from 26.9°/36.7° to 17.3°/20.2°. Orbit voting then brings it to 12.3°/12.3° and the synthetic error to 3.87°. It repairs 62% and 80% of the severe flips on lightbox and sunlamp while breaking only 0.4% of good predictions (Tab. 2). The median errors on real images are 5.3° and 6.3°, so the remaining means are driven by a tail of failures. Orbit distillation improves every setting. The single-view students recover 13–43% of the orbit gain at single-view cost; for example, the PoE greedy error on lightbox drops from 17.3° to 15.9°. Students with orbit voting give the best system: 3.69° synthetic and 11.21°/11.09° on lightbox/sunlamp, about twice the seed std better than the undistilled orbit system, with fewer severe flips. Synthetic validation, the only selection signal, ranks the systems in the same order. MoGe-2 helps on real images: PoE with orbit voting reaches 12.3° on lightbox, against 17.4° for DINOv3-only. It costs about 50 ms per view, half of which is the CPU consensus-mask step. DINOv3-only orbit voting runs at 14 FPS and a single DINOv3 view at 68 FPS (Appendix A).

**Context.** Published SPEED+ results are not directly comparable. Most use detected boxes, full 6-DoF pipelines with PnP or refinement, and fine-tuned networks, whereas we use ground-truth crops and frozen backbones. As reported, keypoint/PnP methods with domain adaptation reach lower real-domain rotation errors (SPNv2: 9.8°/5.6°; Pérez-Villar *et al.*: 2.7°/4.6° on lightbox/sunlamp [4]). Direct regressors are closer to us: SPACE-HOP reports 9.0°/26.6° and FastPose-ViT 28.0°/14.7° [1, 4]. Our contribution is a light frozen-feature design and an analysis of what makes it transfer, not a new state of the art.

### 4.4. Full 6-DoF

Table 3 reports the 6-DoF results. Because our crops come from ground-truth boxes, the box alone already locates the target to 0.55 m. The translation block improves on this by about 7–10× on synthetic data (both pre-registered criteria pass). Joint training does not hurt rotation and even helps slightly on real images: PoE + orbit reaches 3.86°/12.11°/11.73°, against 3.87°/12.30°/12.25°. Orbit averaging lowers translation error by 10–25%. The DINOv3 block is the most accurate on synthetic data (0.059 m), and PoE averaging is best on lightbox (0.175 m). The MoGe-2 block alone is weak for translation. SPACE-HOP reports 0.096/0.302/0.282 m (synthetic/lightbox/sunlamp) [4]. Our numbers are lower, but they rely on ground-truth boxes, use different crop padding and average over 4 views, so the comparison is not matched.

### 4.5. Other datasets

We apply the unchanged recipe to three datasets with per-object trees (Tab. 4). For **T-LESS** [9] we use 7 objects with continuous, discrete or no symmetries, trained on PBR renders and real training images, and tested on the real BOP test targets [10]. For **SYMSOL-T** we use the five shapes with the split of [12]. For **YCB-V** [24] we use 6 objects without symmetry metadata, trained on real training videos only. Labels follow Eq. (1), using the exact roll homography on T-LESS and YCB-V.

*Orbit voting transfers wherever the base model generalises.* It reduces error on T-LESS (12.3° → 9.3° val, 17.9° → 16.6° real test) and on SYMSOL-T (3.8° → 2.6°, with 100% of test predictions within 10°). *PoE fusion is not general.* It passes its pre-registered criterion on T-LESS, but on the textureless SYMSOL-T renders MoGe-2 adds nothing, and on YCB-V it hurts. *YCB-V does not generalise* in this setting. The trees fit their training frames to about 4°, but they memorise the cluttered training videos even though test poses lie within 5° of training poses. Labels, crops and pose coverage were verified (Appendix F). YCB-V training normally adds synthetic renders, which our protocol did not use. *Accuracy is below fine-tuned diffusion.* On our 7 T-LESS objects, LiePose reports R@2/5/10 of 34.9/78.2/88.9%, from its per-object table, whereas we reach 4.0/36.0/73.5%. On SYMSOL-T it reports about 0.4–0.6°, against our 2.6° [12]. Part of this gap is the 2.5° leaf quantisation; part is frozen features and per-object data. LiePose also uses visible-object masks and the full T-LESS test set, so even this comparison is only approximately matched.

## 5. Ablations

All ablations use the PoE variant, 3 seeds and the same evaluation run within each table.

**Hierarchical tree vs. flat grid.** We replace the tree with a count-matched flat classifier over 131,076 Hopf-fibration

| Training | Feat. | Inference | Synth. val | Lightbox | Sunlamp | >150° (%) | | ms |
|---|---|---|---|---|---|---|---|---|
| | | | mean $E_R$ (°), 3 seeds | | | LB | SL | |
| Clean tree (no augmentation) | PoE | 1 view | 5.54 ±0.07 | 26.91 ±0.78 | 36.66 ±0.93 | 5.9 | 7.4 | 64 |
| Augmented tree (base recipe) | DINOv3 | 1 view | 4.84 ±0.05 | 22.55 ±0.73 | 22.00 ±0.70 | 6.0 | 3.2 | 15 |
| Augmented tree (base recipe) | DINOv3 | orbit (4) | 4.21 ±0.04 | 17.40 ±0.68 | 15.27 ±0.64 | 4.8 | 2.0 | 70 |
| + orbit distillation | DINOv3 | 1 view | 4.57 ±0.05 | 21.61 ±0.82 | 20.55 ±0.94 | 5.8 | 3.1 | 15 |
| + orbit distillation | DINOv3 | orbit (4) | 3.90 ±0.03 | 16.43 ±0.75 | 13.64 ±0.57 | 4.5 | 1.6 | 70 |
| Augmented tree (base recipe) | PoE | 1 view | 4.85 ±0.03 | 17.33 ±0.31 | 20.15 ±0.57 | 4.2 | 3.3 | 64 |
| Augmented tree (base recipe) | PoE | orbit (4) | 3.87 ±0.03 | 12.30 ±0.50 | 12.25 ±0.37 | 3.0 | 1.7 | 279 |
| + orbit distillation | PoE | 1 view | 4.56 ±0.03 | 15.90 ±0.53 | 18.33 ±0.66 | 3.7 | 3.0 | 64 |
| + orbit distillation | PoE | orbit (4) | **3.69** ±0.02 | **11.21** ±0.42 | **11.09** ±0.49 | 2.6 | 1.4 | 279 |

Table 1. **SPEED+ rotation.** Mean $E_R$ over 3 seeds (± seed std), fraction of severe flips (>150°), and end-to-end latency at batch 1. The clean tree is trained without augmentation. Each augmented tree is evaluated in the same run as its distilled students. The DINOv3 student is distilled from a DINOv3-only orbit teacher, and the PoE students from the PoE orbit teacher. Inference cost is unchanged by distillation. All rows use ground-truth crops.

| System | Lightbox | | Sunlamp | |
|---|---|---|---|---|
| | fixed | broken | fixed | broken |
| Aug. tree, 1 view | 51% of 401 | 1.0% | 59% of 206 | 1.0% |
| Aug. tree + orbit | 62% of 401 | 0.4% | 80% of 206 | 0.4% |
| Aug. DINOv3 + orbit | 52% of 401 | 1.3% | 74% of 206 | 0.5% |

Table 2. **Flip transitions relative to the clean PoE tree.** *Fixed*: share of the clean tree's >150° errors that become <20° (denominator given). *Broken*: share of its <20° predictions that become >150°.

| Arm | Feat. | $\hat{t}$ | $E_T$ (m) | | | $E_R$ (°), orbit | | |
|---|---|---|---|---|---|---|---|---|
| | | | Syn. | LB | SL | Syn. | LB | SL |
| Box only | – | – | 0.561 | 0.547 | 0.553 | – | – | – |
| Frozen | DINOv3 | 1 view | 0.090 | 0.233 | 0.240 | 4.21 | 17.40 | 15.27 |
| | | orbit | 0.068 | 0.205 | 0.205 | | | |
| Frozen | PoE | 1 view | 0.108 | 0.206 | 0.232 | 3.87 | 12.30 | 12.25 |
| | | orbit | 0.089 | 0.184 | 0.206 | | | |
| Joint | DINOv3 | 1 view | 0.078 | 0.209 | 0.212 | 4.19 | 17.31 | 14.87 |
| | | orbit | **0.059** | 0.185 | **0.183** | | | |
| Joint | PoE | 1 view | 0.111 | 0.196 | 0.221 | 3.86 | 12.11 | 11.73 |
| | | orbit | 0.093 | **0.175** | 0.194 | | | |

Table 3. **6-DoF with the SPACE-HOP translation target.** Rotation uses PoE or DINOv3 orbit voting. Translation ($\hat{t}$) is predicted from one view or as the orbit average; seed std of $E_T \leq 0.005$ m. Rotation errors are those of the same model with orbit voting. *Box only* uses no image: the object centre is placed at the box centre and the depth at the training mean of $U_z$.

| | | T-LESS (7 obj.) | | SYMSOL-T (5) | | YCB-V (6) | |
|---|---|---|---|---|---|---|---|
| Feat. | Inference | val | test | val | test | val | test |
| DINOv3 | 1 view | 12.7 ±0.1 | 18.6 ±0.2 | 3.78 ±0.03 | 3.82 ±0.03 | 64.3 ±11.6 | 71.2 ±7.0 |
| DINOv3 | orbit (4) | 9.9 ±0.1 | 17.9 ±0.4 | 2.61 ±0.00 | 2.62 ±0.01 | 63.0 ±2.4 | 69.4 ±5.2 |
| PoE | 1 view | 12.3 ±0.1 | 17.9 ±0.2 | 3.87 ±0.03 | 3.88 ±0.01 | 73.5 ±6.8 | 81.1 ±7.3 |
| PoE | orbit (4) | 9.3 ±0.1 | 16.6 ±0.2 | 2.65 ±0.01 | 2.67 ±0.00 | 74.1 ±7.4 | 79.9 ±3.2 |

Table 4. **Other datasets** (object-balanced mean error over 3 seeds; symmetry-aware for T-LESS and SYMSOL-T, raw for YCB-V). One tree is trained per object and branch with the unchanged recipe. T-LESS: synthetic PBR validation, real BOP test. SYMSOL-T: LiePose split. YCB-V: held-out real sequences and keyframes.

| Head (params/branch) | Inference | Synth. | LB | SL |
|---|---|---|---|---|
| Tesseract tree (2.4 M) | 1 view | 4.85 ±0.03 | 17.33 ±0.31 | 20.15 ±0.57 |
| Tesseract tree (2.4 M) | orbit | 3.87 ±0.03 | 12.30 ±0.50 | 12.25 ±0.37 |
| Flat Hopf, 131k (36.1 M) | 1 view | 8.95 ±0.06 | 20.56 ±0.70 | 27.16 ±1.04 |
| Flat Hopf, 131k (36.1 M) | orbit | 7.53 ±0.01 | 13.15 ±0.37 | 16.31 ±0.59 |

Table 5. **Tree vs. flat grid.** Same frozen features, augmentation, training budget and orbit inference. The flat head classifies 131,076 Hopf anchors, matching the tree's cell count.

| Selection over 32 candidates | Synth. | LB | SL |
|---|---|---|---|
| 1 view (greedy) | 4.85 ±0.03 | 17.33 ±0.31 | 20.15 ±0.57 |
| max-confidence | 5.51 ±0.03 | 14.63 ±0.42 | 15.92 ±0.15 |
| top-1 vote | 4.37 ±0.03 | 13.24 ±0.42 | 13.38 ±0.57 |
| weighted vote (32) | 3.46 ±0.03 | 12.58 ±0.42 | 12.88 ±0.49 |
| orbit product | 3.87 ±0.03 | 12.30 ±0.50 | 12.25 ±0.37 |

Table 6. **Selection rule over the same 32 orbit candidates.**

anchors [25], as in SPACE-HOP, keeping everything else identical (Tab. 5). The tree is far more precise (3.87° vs. 7.53° synthetic with orbit voting). It is better on real data in all comparisons and uses 15× fewer parameters. Orbit voting helps both heads, and the flat head with orbit voting even has slightly fewer severe flips on lightbox. Precision therefore comes from the tree, while flip repair comes from orbit voting.

| Augmentation | 1 view | | | orbit (4) | | |
|---|---|---|---|---|---|---|
| | Syn. | LB | SL | Syn. | LB | SL |
| none | 5.54 ±0.07 | 26.90 | 36.66 | 4.62 ±0.04 | 20.25 | 24.83 |
| photometric | 5.00 ±0.03 | 17.38 | 19.68 | 4.25 ±0.06 | 12.29 | 12.37 |
| in-plane roll | 4.91 ±0.00 | 24.30 | 31.52 | 3.78 ±0.03 | 18.69 | 20.70 |
| both | 4.85 ±0.03 | 17.33 | 20.15 | 3.87 ±0.03 | 12.30 | 12.25 |

Table 7. **Which augmentation matters.** *None* is the clean tree. The roll uses exact labels (Eq. (1)).

| Training objective | 1 view, DINOv3 | | PoE + orbit | | |
|---|---|---|---|---|---|
| | Syn. | Syn.-rot. | Syn. | LB | SL |
| Base recipe (GT loss) | 4.84 ±0.05 | 9.88 ±0.22 | 3.87 | 12.30 | 12.25 |
| + paired views | 4.96 ±0.06 | 10.04 ±0.21 | 3.94 | 12.38 | 12.45 |
| + leaf-level transported consistency | 4.59 ±0.04 | 9.43 ±0.14 | 3.71 | 12.41 | 11.74 |
| + hierarchical transported consistency | 4.65 ±0.03 | 9.57 ±0.06 | 3.74 | 12.49 | 11.73 |
| + orbit distillation | 4.57 ±0.05 | 9.33 ±0.18 | **3.69** | **11.21** | **11.09** |

Table 8. **Training-time consistency.** Single-view DINOv3 trees on synthetic and rotated synthetic validation (left) and PoE + orbit (right). The consistency losses transport each view's beam distribution into another view's frame via Eq. (1).

**How to select among orbit candidates.** With the same 32 candidates (Tab. 6), picking the most confident one is clearly worse than any form of agreement, and voting among the four views' top-1 predictions is better. Using all candidates helps most. The kernel-weighted vote is best on synthetic data ($3.46°$). The likelihood product of Eq. (2) is best on both real domains, but its advantage over the vote is within about one seed std. Its pre-registered criterion ("product beats voting on synthetic validation") *fails*. We therefore claim multi-view consistency over hierarchical candidates, not a specific scoring rule.

**Photometric vs. rotation augmentation.** Table 7 separates the two augmentation families. Photometric augmentation with orbit voting reaches $12.3°/12.4°$ on real images, the same as the combined recipe. Rotation augmentation alone gives the best synthetic precision ($3.78°$) but leaves real errors at $18.7°/20.7°$. Orbit voting on the clean tree already removes about a quarter of the real error ($26.9° \rightarrow 20.3°$, $36.7° \rightarrow 24.8°$). The real-domain gain therefore comes from *lighting robustness plus orbit consistency*. The exact roll transform is needed for orbit voting itself; as a training augmentation it mainly improves synthetic precision. This corrects our initial hypothesis that rotation augmentation causes the flip repair.

**Can training learn orbit consistency?** We compared four objectives (Tab. 8):

- paired views of the same image (a data control);
- a transported consistency loss on the leaf distribution only, a standard equivariance regulariser;
- a *hierarchical* transported consistency loss at every tree level, our candidate contribution;
- orbit distillation (Sec. 3.5).

The hierarchical loss beats the base recipe and the paired-view control, but not the leaf-only loss. Its pre-registered criterion fails, so we make no hierarchical-transport claim. Both consistency losses mostly improve synthetic precision. Orbit distillation is the only objective that improves the real domains clearly ($11.2°/11.1°$ with orbit voting).

**Further ablations** are reported in the supplementary material:

- the decoder lineage (MLP, GRU and Transformer decoders, hard vs. soft targets, beam search, tangent residuals);
- fusion (single branches, learned feature fusion, weighted PoE);
- finer $16\times16$ tokens;
- continuous flow posteriors and verifiers, discrete flow matching, and oracle headroom;
- zero-shot transfer of the frozen SPEED+ trees, and the timing breakdown.

## 6. Limitations

- **Ground-truth crops.** All results use ground-truth boxes. For translation in particular, the box alone gives 0.55 m, so a deployable claim needs a detector or perturbed boxes.
- **Comparisons.** External SPEED+ numbers are not matched comparisons.
- **Quantisation floor.** The depth-5 leaf quantisation ($2.51°$ mean) bounds fine accuracy, and our best synthetic errors ($3.3$–$3.7°$) are close to it. A continuous refinement on top of the selected mode is a natural next step; the refiners we tried helped only on synthetic data.
- **Cost.** Orbit voting costs $4\times$ inference.
- **Scope of the components.** MoGe-2 fusion does not help on every dataset. YCB-V trained on real videos alone does not generalise, and on T-LESS and SYMSOL-T we remain behind fine-tuned diffusion models.
- **Timing.** Timings come from a shared GPU and are upper bounds.

## 7. Conclusion

A small tesseract tree on frozen DINOv3 and MoGe-2 features, fused by a training-free product of experts, is an accurate rotation predictor on synthetic spacecraft images. Its failures on real images are mostly flips to a look-alike mode that the model already considers. Scoring hierarchical hypotheses for consistency across exactly mapped in-plane rotations repairs most of these flips, and distilling that consistency into single-view students improves the result further,

to $11.2°/11.1°$ on the SPEED+ real domains. Pre-registered ablations attribute the gains to the tree (precision), to photometric augmentation and orbit voting (real-domain flips), and to distillation (both), and they rule out several plausible alternatives. Our evaluation is intentionally conservative. We hope the protocol and the negative results are as useful as the method.

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# Orbit-Consistent Tesseract Trees on Frozen Foundation Features for Spacecraft Pose Estimation across the Domain Gap

Supplementary Material

## A. Timing

| System | batch 1 | | batch 64 | |
|---|---|---|---|---|
| | ms | FPS | ms/img | img/s |
| DINOv3, 1 view | 14.7 | 68.1 | 2.9 | 347 |
| DINOv3 + orbit | 69.6 | 14.4 | 12.1 | 82 |
| PoE, 1 view | 64.0 | 15.6 | 50.0 | 20 |
| PoE + orbit | 278.5 | 3.6 | 206.8 | 5 |

Table 9. **End-to-end latency** of the augmented trees (shared RTX 6000 Ada; upper bounds; image already decoded).

At batch 1 (Tab. 9), a DINOv3 forward pass takes 6.4 ms and one tree 7.4 ms. MoGe-2 adds 21 ms for the network and 20 ms for the CPU consensus-mask step per view, which is the main optimisation target. Orbit voting multiplies the cost by 4.4–4.7×. Batched evaluation reduces tree cost to 0.1–0.3 ms per image.

## B. Model lineage and fusion

| Design (clean trees, PoE greedy) | Synth. | LB | SL |
|---|---|---|---|
| MLP path decoder, hard targets | 6.24 | 31.1 | 43.7 |
| GRU path decoder | 6.15 | – | 40.6 |
| MLP decoder, geodesic soft targets | 6.07 | 28.2 | 42.1 |
| Transformer decoder, hard targets | 5.69 | 31.6 | 41.8 |
| **Transformer + soft targets** | **5.54** | **26.9** | **36.7** |
| + beam 8 instead of greedy | no gain | – | – |
| + tangent residual at the leaf | −0.13 | ±0.0 | ±0.0 |
| 16×16 instead of 8×8 DINOv3 tokens | 5.56 | 27.0 | 36.0 |
| DINOv3 tree alone | 5.35 | 37.1 | – |
| MoGe-2 tree alone | – | 43.7 | – |
| Learned concat fusion, one Transformer (8×8) | 5.34 | 40.8 | – |
| Equal per-level PoE (same grid) | 5.58 | 27.9 | – |
| Weighted PoE, $w_{\text{DINO}}$=0.8 (proxy-selected) | 5.14 | 32.0 | 36.7 |

Table 10. **Design steps** on clean (non-augmented) trees, seed means. Dashes: not evaluated in that study.

Table 10 lists the design steps that led to the base tree. The Transformer decoder with geodesic soft targets was best; beam search at full data brought no gain, and a tangent residual only refined errors within the correct cell. Fusion is strongly asymmetric across domains. A learned fusion that feeds concatenated tokens into one Transformer matches DINOv3 alone on synthetic data but is much worse on real images (40.8° vs. 27.9° for PoE on the same grid). We interpret this as the learned fusion over-fitting render statistics, while training-free PoE keeps the branches' errors partly independent. A weighted PoE selected on the rotated validation proxy ($w$=0.8) is 5° worse on lightbox than equal weights. Following our protocol, we did not use real data to choose another weight. The consensus foreground mask reduces MoGe-2's background leakage on real images from 0.52 to 0.12 (lightbox) and from 0.57 to 0.12 (sunlamp). It is what makes the MoGe-2 branch useful.

## C. Negative results

| Approach (on clean trees unless noted) | Synth. | LB | SL |
|---|---|---|---|
| Clean PoE tree (reference) | 5.54 | 26.9 | 36.7 |
| $SO(3)$ flow posterior on tree memory, KDE mode | 5.22 | 29.8 | 37.6 |
| best of 32 samples (*oracle*) | 2.60 | 9.7 | 10.6 |
| Flow re-noise verifier over beams + flips + corners | 4.78 | 26.5 | 32.7 |
| best candidate (*oracle*) | 2.25 | 7.9 | 11.4 |
| Kurz corner tree [14] | 5.20 | 28.8 | 39.1 |
| Tree distilled from flow masses ($\beta$=1, 1 seed) | 5.62 | 26.2 | 37.7 |
| Discrete flow matching on tree paths | 7.22 | 29.8 | 39.8 |
| Aug. DINOv3 tree (reference) | 4.84 | 22.6 | 22.1 |
| + $SO(3)$ flow, KDE | 4.10 | 25.2 | 24.4 |

Table 11. **Refiners and verifiers that did not fix real flips** (seed means). Oracles are not deployable and show headroom only.

Table 11 collects approaches that improved synthetic accuracy but not, or only slightly, real accuracy. A Riemannian flow-matching posterior [2] conditioned on the tree's encoder memory contains good samples: its best-of-32 oracle reaches 9.7° on lightbox. It also assigns the ground truth a much higher log-likelihood than the tree does (3.85 vs. 0.98 nats on lightbox), and its 90% regions over-cover (0.98). Yet its mode estimate is worse than the tree's. Re-scoring an expanded candidate set (beam, 180° flips, cell corners) by re-noise/return consistency helps synthetic data and sunlamp, but does not repair lightbox flips. Synthetic render-and-compare templates also failed to choose the right flip. A discrete flow on the tesseract path [5] was clearly worse. We note that interpolating path tokens is not geometry-preserving: 45–53% of intermediate paths lie more than 30° from both endpoints. Distilling flow masses into the tree did not pass its criterion. An oracle that stops decoding at the best depth would gain only 0.51° on synthetic data, so adaptive depth was not pursued. These results support the diagnosis of Sec. 4.2: the right mode is reachable, but scorers trained on clean synthetic features cannot select it on real images. Lighting-robust training and multi-

view consistency can.

## D. Orbit distillation and 6-DoF details

Retention of the orbit gain by the single-view students, defined as (base greedy − student greedy)/(base greedy − base orbit), is 43%/18%/22% (DINOv3) and 30%/28%/23% (PoE) on synthetic/lightbox/sunlamp, and 13–15% on the rotated proxy. For the joint 6-DoF model, the ESA-style score (rotation error in radians plus normalised translation error, with the challenge precision floors) is 0.084/0.241/0.238 for PoE.

## E. Per-object results

| Object | PoE+orbit | DINOv3+orbit |
|---|---|---|
| *T-LESS* | | |
| 1 | 23.82 | 25.00 |
| 4 | 14.50 | 15.69 |
| 5 | 6.22 | 7.07 |
| 11 | 11.90 | 12.92 |
| 18 | 14.38 | 14.78 |
| 21 | 34.78 | 37.24 |
| 27 | 10.36 | 12.43 |
| *SYMSOL-T* | | |
| cone | 1.83 | 1.79 |
| cube | 2.72 | 2.90 |
| cyl | 1.67 | 1.65 |
| icosa | 3.81 | 3.71 |
| tet | 3.35 | 3.06 |
| *YCB-V* | | |
| 003_cracker_box | 85.27 | 74.61 |
| 006_mustard_bottle | 97.96 | 87.29 |
| 011_banana | 61.63 | 83.23 |
| 025_mug | 24.25 | 22.76 |
| 035_power_drill | 41.34 | 25.07 |
| 037_scissors | 169.07 | 123.72 |

Table 12. **Per-object test error** (3-seed mean; symmetry-aware for T-LESS and SYMSOL-T).

On T-LESS (Tab. 12), most of the test error comes from object 21 (no symmetry metadata, near-symmetric; 18% of predictions above 90°) and object 1 (continuous symmetry). Continuous symmetries are evaluated on a 10°-step discretisation of the symmetry set, which adds up to 5° of artificial error for objects 1 and 4 and for the cone and cylinder.

## F. YCB-V diagnosis

On YCB-V we checked four things:

1. The trees fit their training frames: 3.8° and 3.9° on mustard bottle and mug.
2. Test poses are covered by training: the median nearest training rotation is about 5° away for cracker box, mustard bottle and banana, whose test errors are still 75–87°.
3. Every crop contains all of the object's label pixels.
4. The ground-truth rotations agree with the segmentation: 65–98% of the projected model points fall on the object mask, against 28–83% for a transposed-rotation control.

The failure is therefore memorisation of the 16–20 cluttered training videos per object. The object covers only 5–40% of the crop and the DINOv3 tokens are not masked. This is not a label or pipeline error. Standard YCB-V training adds synthetic renders, which we did not use.

## G. Zero-shot transfer of the frozen SPEED+ trees

| Dataset | Raw $E_R$ | | Frame-free pair error | | |
|---|---|---|---|---|---|
| | frozen | chance | frozen | chance | constant |
| T-LESS | 117.5 | 103.6 | 95.4 ±3.0 | 104.5 | 88.7 |
| SYMSOL-T | 49.4 | 53.7 | 47.8 ±0.5 | 53.7 | 53.8 |
| YCB-V | 115.6 | 125.8 | 101.6 ±2.8 | 125.8 | 57.8 |

Table 13. **Zero-shot**: frozen SPEED+ PoE trees with orbit voting, no retraining, test split. The pair error $\min_S \angle(\hat{R}_i \hat{R}_j^\top, R_i S R_j^\top)$ over random same-object pairs cancels the unknown offset between object frames. *Chance*: uniformly random predictions. *Constant*: one rotation predicted for every image.

Object frames are arbitrary, so raw errors are expected near chance. The frame-free pair error (Tab. 13) is therefore the meaningful measure. On T-LESS and YCB-V it is *worse* than predicting one constant rotation, because both datasets have strong pose priors. On SYMSOL-T, where rotations are uniform, the trees beat chance only because of one shape: the cube (27.8° vs. 41.2°), plausibly because the spacecraft body is box-like. There is no general zero-shot transfer: the trees must be retrained per object.

## H. Disclosures and reproducibility

- **Late feature-equivalence check.** The augmented and rotated views are extracted by a compact feature extractor. A pre-registered check against the full feature caches was run after the main results. It failed its strict element-wise tolerance: the maximum standardised difference was 0.51 for MoGe-2 features (p99 0.004; correlation 0.999998). Its functional follow-up showed that no PoE prediction changed on 512 crops, and that single-branch greedy leaves changed for 1/512 (DINOv3) and 3/512 (MoGe-2) crops.
- **SYMSOL-T bug.** Nine MoGe-2 SYMSOL-T trees first trained to NaN: always-empty border cells had a standard deviation near $10^{-6}$, and float16 overflowed. They were retrained with a standard-deviation floor of $10^{-2}$; all other trees were unaffected. Only the corrected evaluation is reported.
- **YCB-V paths.** A file-path error in the YCB-V manifest (colour images are JPEG) was fixed before any YCB-V

result.

- **Post-hoc control.** The constant-prediction control in Tab. 13 was added after the T-LESS zero-shot result and is labelled as post hoc.

Every number in the results tables is generated by a script directly from the evaluation logs of the named runs.